Reference-Aware Selective Prediction for Frozen Visual Prompting Pipelines

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Abstract

Visual prompting pipelines—a frozen matcher that feeds point prompts to a frozen segmenter—have become a standard approach for few-shot segmentation. However, they lack a mechanism to assess whether a given reference image will produce a successful segmentation. We formalize this as the **Reference Reliability Estimation** (RRE) problem: given a query and a set of candidate references, predict whether each reference will succeed. On MoNuSAC, swapping the reference image can change the Dice score from 0.82 to 0.02, and over 14% of queries experience catastrophic failure (mean Dice across references < 0.1) from at least one reference. We systematically characterize two reference-induced failure modes—instability (large Dice variance across references) and catastrophic failure—and find that they exhibit a reproducible empirical association in SAM-B pipelines (Pearson $r = -0.51$, $p < 0.001$; partial $r = -0.38$ controlling for mean Dice). However, this association does not generalize to SAM2 pipelines (SAM2-Tiny: $r = -0.11$, $p = 0.33$; SAM2-Small: $r = -0.18$, $p = 0.11$), qualifying the scope of the phenomenon. We propose RAMP (Reference-Adaptive Multi-Prompt Fusion), a post-hoc selective predictor that uses a lightweight supervised risk estimator (gradient boosting) with frozen foundation models. RAMP reduces the catastrophic failure rate by 37.5–84.4% at 80% coverage on MoNuSAC across four pipeline configurations, with minimal Dice loss ($< 1.1\%$). We establish a reference selection benchmark evaluating five strategies across MoNuSAC, and we validate the approach on IDRID and MoNuSeg. A simple frozen top-K pipeline (DINOv2-L + SAM-B) achieves Dice = 0.446 on MoNuSAC, outperforming five recent methods under shared reference-resampling protocol, demonstrating that reference reliability is a dimension orthogonal to prompt sophistication. All results are reported at the query level with bootstrap confidence intervals and Holm-corrected significance tests.

1 Introduction

A user performing few-shot segmentation faces a decision that the literature has largely ignored: which reference image should they choose? The answer matters more than most practitioners realize. On MoNuSAC, a query image can obtain Dice = 0.82 with one reference and Dice = 0.02 with another. The pipeline provides no mechanism to distinguish these outcomes. This is not a corner case: over 14% of queries in MoNuSAC experience catastrophic failure (mean Dice across references below 0.1) from at least one reference. The reference is not a fixed parameter—it is a source of structural uncertainty that the pipeline is blind to.

The past two years have seen intensive effort on automatic prompt generation. GF-SAM [1] (NeurIPS 2024 Spotlight) uses graph-based point clustering; MAUP [2] (MICCAI 2025) performs K-means on confidence maps; SynPo [3] (MICCAI 2025) fuses DINOv2 and SAM confidence; Matcher [4] (ICLR 2024) uses bidirectional patch matching; PerSAM [5] (ICLR 2024) injects target-guided attention; CycleSAM [6] (CVAI 2024) enforces spatial cycle-consistency; H-SAM [7] (TPAMI 2025) extends SAM hierarchically; and SAM-Med2D [8] adapts SAM for medical images. All optimize *how* to construct prompts from a given reference. None addresses *whether* the reference is reliable.

This paper identifies a missing component in the visual prompting paradigm: **Reference Reliability Estimation (RRE)**. RRE asks: given a query and a candidate reference, what is the probability that the ref-

erence will produce a successful segmentation? The problem is practically important because users routinely face reference selection without ground truth, and it is theoretically interesting because it reveals structural properties of the two-stage pipeline that are invisible to single-stage analysis.

The central insight of this paper is that reference-induced failure has *structure*. Instability (large Dice variance across references) and catastrophic failure (mean Dice < 0.1) are not independent: they exhibit a reproducible empirical association. A query that produces highly variable results across references rarely fails catastrophically; a query that fails catastrophically shows almost no variance across references. This pattern is significant in SAM-B pipelines (DINOv2-L+SAM-B: Pearson $r = -0.51$, $p < 0.001$; DINOv2-B+SAM-B: $r = -0.55$, $p < 0.001$; both significant after Holm correction). However, the association is weaker and non-significant in SAM2 pipelines (SAM2-Tiny: $r = -0.11$, $p = 0.33$; SAM2-Small: $r = -0.18$, $p = 0.11$), and the partial correlation controlling for mean Dice is positive for SAM2-Tiny ($r_{\text{partial}} = +0.13$). Together, these results indicate that the anti-correlation is a property of specific pipeline architectures rather than a universal law.

We trace why this association exists through the geometry of the point-prompt bottleneck (Section 3). The key quantity is the concentration of the matching map produced by the matcher. When the matching map has sharp peaks, point coordinates are tightly clustered around the target, and changing the reference shifts the peaks slightly while keeping points on-target—producing high instability but low catastrophic risk. When the matching map is diffuse, points are scattered, and reference changes have negligible effect on already-random coordinates—producing low instability but high catastrophic risk. We formalize this mechanism through three geometric hypotheses linking matching-map concentration to prompt dispersion to catastrophic risk, and we provide empirical verification with validity controls to rule out metric coupling artifacts (Section 3.4).

We address the RRE problem with **RAMP (Reference-Adaptive Multi-Prompt Fusion)**, a post-hoc selective predictor. RAMP uses a lightweight supervised risk estimator (gradient boosting) with four test-time features—matching-map disagreement, candidate Jaccard, prompt overlap, and prediction consistency—to estimate catastrophic risk per reference. It then weights references by predicted reliability and fuses predictions via risk-stratified consensus. The result is a selective predictor that reduces the catastrophic failure rate by 37.5–84.4% at 80% coverage on MoNuSAC, using coverage-based ranking that does not require a fixed probability threshold. The risk estimator is trained on labeled multi-reference data but uses frozen matcher and segmenter models.

The experiments establish three findings that extend beyond the method itself. First, a simple frozen top-K pipeline (DINOv2-L + SAM-B) achieves Dice = 0.446 on MoNuSAC, outperforming GF-SAM (SAM-H, 0.388), MAUP (SAM-H, 0.279), CycleSAM (0.412), H-SAM (0.405), and SAM-Med2D (0.438) under a shared reference-resampling protocol. We note that the backbones differ (GF-SAM and MAUP use SAM-H, while our pipeline uses SAM-B), so the comparison illustrates the importance of reference reliability rather than claiming method superiority. Second, we establish a reference selection benchmark evaluating five strategies (random, similarity, entropy, oracle, and a learned estimator) on MoNuSAC, with cross-dataset validation on IDRID and MoNuSeg. Third, we characterize the conditions under which selective prediction is effective, identifying the catastrophic prevalence regime where the method provides benefit.

The paper's contributions are threefold. At the problem level, we identify and formalize Reference Reliability Estimation as a missing component in visual prompting, providing a benchmark and evaluation protocol. At the analytical level, we characterize the empirical association between instability and catastrophic failure through geometric hypotheses and validity-controlled experiments, while qualifying the scope of this association across pipeline architectures. At the method level, we provide RAMP, a post-hoc selective predictor with a supervised risk estimator that achieves 37.5–84.4% CFR reduction with frozen foundation models.

2 Related Work

2.1 Visual Prompting and Prompt Generation

The SAM family [9, 10] enables segmentation through spatial prompts without task-specific training. This capability has spurred a wave of automatic prompt generation methods, all sharing the same two-stage structure: matcher extracts features, computes correspondence, and converts to prompts. The methods differ in the prompt conversion strategy. GF-SAM [1] uses graph centrality with PNA and PMC clustering. MAUP [2] performs adaptive K-means on confidence maps. SynPo [3] fuses DINOv2 and SAM confidence with anatomical constraints. Matcher [4] uses bidirectional patch matching with instance-level merging. PerSAM [5] injects target-guided attention into SAM's decoder. CycleSAM [6], H-SAM [7], and SAM-Med2D [8] extend the paradigm for surgical, document, and medical domains respectively.

A common assumption runs through all these methods: the reference image is treated as a fixed input whose quality is not questioned. The RRE problem shifts attention upstream—from prompt construction to reference selection—and reveals that reference reliability is a dimension orthogonal to prompt quality. Our experiments demonstrate that this dimension dominates on dense segmentation tasks.

2.2 Selective Prediction and Uncertainty

Selective prediction [11, 12] allows models to defer decisions on uncertain inputs. Standard approaches estimate uncertainty through MC Dropout [13] or Deep Ensembles [14], operating on model parameters. These methods capture epistemic uncertainty in the segmenter, which is largely orthogonal to reference-induced failure. Our approach differs in both the source of uncertainty signals (pipeline-level features from multi-reference runs) and the use of ranking-based selection (coverage-based ranking avoids per-model calibration, enabling cross-dataset transfer).

2.3 Reference-Based Few-Shot Learning

The idea of selecting or weighting references has been explored in metric-based few-shot learning [15, 16], where support set quality affects performance. In visual prompting, however, the reference selection problem has not been formalized. The closest work is in cross-domain few-shot segmentation [17], where domain gap between reference and query is addressed through feature adaptation rather than reliability estimation. Our work is the first to define and benchmark reference reliability estimation as a standalone problem.

2.4 Scale and Reliability

The relationship between model scale and reliability is not always monotonic [18, 19]. Our factorial audit provides a controlled case: SAM-L (312M) achieves higher mean Dice than SAM-B (94M) but shows similar catastrophic failure rates, and the anti-correlation weakens substantially (from $r = -0.51$ to $r = -0.22$). Larger models are more stable but not necessarily more reliable in the reference-induced failure sense.

3 The Point-Prompt Bottleneck: Mechanism Analysis

The failure modes of the two-stage visual prompting pipeline—reference-induced instability and catastrophic failure—exhibit a reproducible empirical association in SAM-B pipelines. This section defines the key quantities, proposes three geometric hypotheses about the failure mechanism, and derives testable predictions that guide both the feature design and the experimental validation.

3.1 Preliminaries

Consider a query image q and a set of references $\{r_i\}_{i=1}^{K}$. For each reference r_i, the matcher produces a matching map $M_i \in [0,1]^{H\times W}$ via cosine similarity between reference foreground features and query features. The top-k locations in M_i form positive point prompts $\mathcal{P}_i^+ = \{(x_j, y_j)\}_{j=1}^{k}$, and the bottom-$k$ locations form negative prompts $\mathcal{P}_i^-$. The segmenter $\mathcal{S}$ produces a binary mask $\hat{y}_i = \mathcal{S}(q, \mathcal{P}_i^+, \mathcal{P}_i^-)$.

We define the following quantities. The matching map concentration is measured via the normalized entropy $H(M_i) = -\sum_{h,w} M_i(h,w) \log M_i(h,w)$. The prompt dispersion σ_i^2 is the variance of the top-k point coordinates around their centroid. The point localization error PLE_i is the mean Euclidean distance from the top-k points to the nearest ground-truth target pixel. The matching map bias d_i^* is the distance from the matching map peak to the nearest target pixel.

3.2 Geometric Hypotheses

We propose three geometric hypotheses about the failure mechanism. These hypotheses are not presented as formal theorems but as empirically testable propositions that organize the experimental investigation.

Hypothesis H1 (Entropy–dispersion association). *Matching map concentration is associated with prompt spatial dispersion: when the matching map has sharp, well-localized peaks, the top-k points tend to be tightly clustered; when the matching map is diffuse, the points tend to be scattered.*
Geometric intuition. The top-k selection operator samples locations from the matching map distribution. When probability mass is concentrated, repeated sampling yields similar coordinates; when mass is dispersed, sampling yields diverse coordinates. We do not claim that entropy uniquely determines dispersion—two maps with identical entropy but different spatial arrangements can produce different dispersions. The association is statistical, not deterministic.

Hypothesis H2 (Localization–catastrophe association). *When prompts are systematically mispositioned relative to the target, catastrophic failure is more likely. The severity depends on both the dispersion (random scatter) and the bias (systematic offset) of the prompt coordinates.*
Geometric intuition. A point prompt can only guide the segmenter toward the target if it is placed sufficiently close. The triangle inequality bounds the localization error by the sum of dispersion and bias: $\text{PLE}_i \leq \sigma_i + d_i^*$. When both terms are small, the prompt reliably targets the object; when either term is large, the prompt may miss the target entirely.

Hypothesis H3 (Dissociated failure signatures). *Reference-induced instability and catastrophic failure carry dissociated information because they originate from different pipeline levels: instability is dominated by matching-map-level variation (which reference is chosen), while catastrophic failure is dominated by mask-level agreement (whether any reference can produce a valid mask).*
Geometric intuition. Instability arises from the sensitivity of the matching map peak to the reference choice: small perturbations in the reference shift the peak location, changing d_i^* and thus the Dice score. Catastrophic failure arises when the matching map is either too diffuse (H1) or too biased (H2) for any reference to place prompts correctly. These two failure pathways operate at different stages of the pipeline and are therefore driven by different features.

3.3 Empirical Mechanism Analysis

We test the three hypotheses on MoNuSAC using a reference-resampling protocol with $K = 6$ patient-disjoint references per query, aggregated at the query level ($N = 80$ queries) across seeds and prompt configurations.

H1: Entropy–dispersion association. The entropy proxy (matching map disagreement across references) correlates with prompt spatial overlap at Pearson $r = 0.48$ ($p < 0.001$), confirming that diffuse matching maps produce more scattered prompts. The entropy-instability relationship is also significant ($r = -0.45$, $p < 0.001$): as entropy increases, the best-worst Dice gap across references decreases, consistent with the prediction that diffuse maps are insensitive to reference choice.

H2: The non-monotonic entropy–CFR relationship. Hypothesis H2 predicts that catastrophic risk increases with localization error. The entropy–CFR relationship is non-monotonic: the CFR is highest in the mid-entropy regime (entropy bins 0.4–0.6) and lower in both the low-entropy and high-entropy regimes. This finding is consistent with the geometric intuition: low-entropy maps produce precise prompts (low catastrophic risk), high-entropy maps produce scattered but potentially target-spanning prompts (moderate risk), and mid-entropy maps produce prompts that are too precise to explore the full target but too imprecise to reliably hit it (highest risk). This non-monotonicity is an important qualification: it means that simple entropy-thresholding cannot distinguish between low-risk and high-risk queries.

H3: Feature dissociation. The features that predict instability (matching-map disagreement, prompt overlap) are weakly correlated with the features that predict catastrophic failure (prediction consistency, mask Jaccard), with cross-feature Pearson $r < 0.3$ in most cases. This dissociation supports the design choice of using separate risk signals for the two failure modes.

3.4 Validity Controls: Ruling Out Metric Coupling

A potential concern is that the observed association between instability ($\max$ Dice $-$ $\min$ Dice) and catastrophic failure (mean Dice < 0.1) is partly a metric artifact: when all reference Dice scores are low, both instability and CFR are mechanically constrained. We conduct four control experiments to assess this concern.

Control A (Split-reference independence). References are partitioned into two disjoint groups (A = refs 1–3, B = refs 4–6). Instability computed from Group A is correlated with catastrophic failure from Group B, and vice versa. The split-reference correlation ($r = -0.51$, 95% CI $[-0.55, -0.44]$) is weaker than the coupled correlation ($r = -0.62$) but remains significant and directionally consistent, indicating that the association is not purely a metric artifact.

Control B (GT-free instability). We compute instability using prediction-mask disagreement (Jaccard distance between reference masks) instead of Dice-based instability. The GT-free instability still correlates with catastrophic failure at $r = -0.59$ ($p < 10^{-8}$), confirming that the association persists when the instability metric is definitionally independent of the failure metric.

Control C (Null simulation). The observed anti-correlation ($r = -0.62$) is compared against three null models. Null 1 (random beta distribution preserving per-query mean and standard deviation) produces $r = -0.58$, indicating that a substantial portion of the anti-correlation can be reproduced by chance when the marginal distribution is preserved. However, Null 2 (shuffled query-reference structure, $r = +0.64$) and Null 3 (independent uniform Dice, $r = +0.38$) produce correlations in the opposite direction, confirming that the observed negative correlation is not a generic property of the metric definitions but reflects genuine structure in the data. The structural signal beyond Null 1 is $\Delta r = -0.036$.

Control D (Partial correlation). After controlling for mean Dice (query difficulty), the partial correlation between instability and CFR is $r_{\text{partial}} = -0.25$ (95% CI $[-0.41, -0.07]$), which excludes zero. This indicates that even after accounting for overall difficulty, queries with higher reference-induced instability tend to have lower catastrophic failure rates.

Summary. The validity controls confirm that the instability-CFR association has a genuine structural component beyond metric coupling. However, the association is weaker than the raw correlation suggests, and

Null 1 shows that a substantial portion can be explained by the marginal distribution. We therefore characterize the finding as a *reproducible empirical association* rather than a structural theorem. Furthermore, the query-level reanalysis (Section 6) shows that the association is significant only in SAM-B pipelines and not in SAM2 pipelines, qualifying the scope of the phenomenon.

4 Method: RAMP

4.1 Problem Formulation

Given a query image q and a set of K reference images $\{r_i\}_{i=1}^{K}$, the visual prompting pipeline produces K segmentation masks $\{\hat{y}_i\}_{i=1}^{K}$. The RRE problem asks: for each reference r_i, what is the probability that the resulting mask $\hat{y}_i$ achieves a successful segmentation (Dice ≥ 0.5)? The selective prediction problem asks: can we fuse the available masks into a single prediction that is reliable, and if not, can we abstain?

4.2 Post-hoc Selective Prediction

RAMP operates as a post-hoc selective predictor. The pipeline processes all K references through the frozen matcher and segmenter, extracts pipeline-level features from the outputs, and then applies a trained risk estimator to decide whether to accept or abstain. This design avoids the circular dependency that would arise from using risk estimates to configure prompts before running the segmenter.

4.2.1 Feature Extraction

For each reference r_i, we extract four pipeline-level features:

1. **Matching-map disagreement** (f_1): the mean Jaccard distance between the matching map of r_i and the matching maps of the other references. This captures the entropy of the matching map without requiring ground truth.

2. **Candidate Jaccard** (f_2): the mean Jaccard similarity between the top-k candidate regions of r_i and those of the other references. This captures the consistency of the matcher's region proposals.

3. **Prompt overlap** (f_3): the mean overlap between the prompt coordinates of r_i and those of the other references. This captures the spatial dispersion of the prompts.

4. **Prediction consistency** (f_4): the mean Dice similarity between the mask $\hat{y}_i$ and the masks from other references. This captures whether the segmenter produces consistent outputs across references.

Features f_1–f_3 are computed from matcher outputs and do not require the segmenter. Feature f_4 requires all K masks and is computed after the segmenter pass. All features are computed without ground truth.

4.2.2 Risk Estimation

We train a gradient-boosted tree classifier (100 trees, max depth 3) to predict whether a reference will produce a catastrophic failure (Dice < 0.1). The classifier takes the four features as input and outputs a catastrophic risk score $c_i \in [0, 1]$ for each reference. The training uses labeled multi-reference data from the MoNuSAC training set. The matcher and segmenter remain frozen throughout.

Algorithm 1 RAMP: Post-hoc Selective Prediction

Require: Query q, references $\{\mathbf{r}_i\}_{i=1}^{K}$, trained risk predictor
Ensure: Selective prediction decision and fused mask

for each reference $\mathbf{r}_i$ **do**
    Run matcher: $\mathbf{M}_i \leftarrow \text{Matcher}(q, \mathbf{r}_i)$
    Compute f_1, f_2, f_3 from $\mathbf{M}_i$ and other $\mathbf{M}_j$
    Run segmenter: $\hat{\mathbf{y}}_i \leftarrow \text{Segmenter}(q, \text{TopK}(\mathbf{M}_i))$
end for
for each reference $\mathbf{r}_i$ **do**
    Compute f_4 from $\{\hat{\mathbf{y}}_j\}_{j=1}^{K}$
    $\mathbf{f}_i \leftarrow (f_1, f_2, f_3, f_4)$
    $c_i \leftarrow \text{RiskPredictor}(\mathbf{f}_i)$
end for
$\bar{c} \leftarrow \frac{1}{K}\sum_i c_i$
$w_i \leftarrow \exp(-c_i/\tau) \;/\; \sum_j \exp(-c_j/\tau)$
$\mathbf{m}_{\text{fused}} \leftarrow \text{RiskStratifiedConsensus}(\{\hat{\mathbf{y}}_i\}, \{w_i\}, \bar{c})$
if rank-based coverage condition satisfied **then**
    return $\mathbf{m}_{\text{fused}}$
else
    return ABSTAIN
end if

4.2.3 Reference Weighting and Fusion

Given the risk scores $\{c_i\}_{i=1}^{K}$, we compute reference weights via softmax with temperature τ:

$$w_i = \frac{\exp(-c_i/\tau)}{\sum_{j=1}^{K} \exp(-c_j/\tau)} \tag{1}$$

The fused mask is obtained via risk-stratified consensus: if the mean risk $\bar{c} = \frac{1}{K}\sum_i c_i$ is below a threshold, we use weighted majority voting; if $\bar{c}$ is above the threshold, we use only the highest-weight reference's mask. The threshold is set to 0.3 based on the training data.

4.2.4 Coverage-Based Abstention

Rather than using a fixed probability threshold for abstention, RAMP uses coverage-based ranking. Query-reference pairs are ranked by their predicted risk, and the top $C\%$ (by coverage) are selected for prediction. This ranking-based approach does not require a calibrated probability threshold and transfers across datasets without recalibration. We report results at 80% coverage (i.e., the 20% of queries with the highest predicted risk are deferred).

4.3 Computational Cost

With $K = 6$ references, each query requires $K + 1 = 7$ matcher calls (query + 6 references) and $K = 6$ SAM calls. The risk predictor inference adds negligible cost (< 1 ms). The total cost is approximately $7\times$ the cost of a single-reference pipeline, which is the standard cost of any multi-reference approach. Feature extraction is the main overhead; f_4 (prediction consistency) requires all K masks and is the most expensive feature to compute.

5 Experimental Setup

5.1 Datasets and Reference Protocol

We evaluate on three medical image segmentation datasets. **MoNuSAC** (80 holdout queries, 6 patient-disjoint references per query) is the primary dataset for method development and ablation. **MoNuSeg** (30 queries) and **IDRID** (30 queries) serve as secondary validation datasets. All datasets use the same reference-resampling protocol: for each query, $K = 6$ references are sampled from the training set with patient-level disjointness, and results are aggregated across 5 random seeds and 3 prompt configurations (P20-N1, P30-N0, P50-N0). The primary statistical unit is the query ($N = 80$ for MoNuSAC), with within-query aggregation over seeds and configurations.

We also report initial results on natural image benchmarks (Pascal-5^i and COCO-20^i, fold-0 only) to assess cross-domain generality. Full 4-fold evaluation is planned for future work.

5.2 Pipeline Configurations

We evaluate four pipeline configurations: DINOv2-L + SAM-B (primary), DINOv2-L + SAM-L, DINOv2-L + SAM2-Tiny, and DINOv2-B + SAM-B. The matcher (DINOv2) and segmenter (SAM/SAM2) are frozen throughout all experiments.

5.3 Baselines

Reference selection baselines (5 strategies): random selection, DINOv2 feature similarity ranking, matching-map entropy ranking, oracle selection (best reference by ground-truth Dice), and a learned reliability estimator (gradient-boosted trees, described in Appendix A).

External methods: GF-SAM [1], MAUP [2], CycleSAM [6], H-SAM [7], and SAM-Med2D [8]. We note backbone differences: GF-SAM and MAUP use SAM-H, while our pipeline uses SAM-B. The comparison illustrates the importance of reference reliability rather than claiming method superiority under identical conditions.

Uncertainty baselines: MC Dropout (10 forward passes, dropout rate 0.1) and Deep Ensemble (SAM-B + SAM-L + SAM2-Tiny).

5.4 Evaluation Metrics

Primary: Catastrophic failure rate (CFR) at 80% coverage, defined as the proportion of accepted queries with mean Dice < 0.1. **Secondary**: CFR reduction (relative to the uncorrected baseline), mean Dice at 80% coverage, AURC (Area Under Risk-Coverage curve), and excess AURC (relative to random baseline). **Statistical reporting**: All metrics are aggregated at the query level. Paired comparisons use Wilcoxon signed-rank tests with Holm correction. Bootstrap 95% confidence intervals (10,000 resamples) are reported for all primary metrics.

6 Results

6.1 Query-Level Anti-Correlation Analysis

Table 1 reports the query-level anti-correlation between instability and CFR across all five pipeline configurations. After Holm correction, only the two SAM-B pipelines show significant anti-correlation. The partial correlation controlling for mean Dice remains significant for SAM-B pipelines but is negligible for SAM-L

Table 1: Query-level anti-correlation between instability and CFR. $N = 80$ queries. Pearson r with Holm correction. Partial r controls for mean Dice. Bold: $p < 0.05$ after Holm correction.

Pipeline	Pearson r	p (Holm)	Partial r	Partial p
DINOv2-L + SAM-B	-0.512	< 0.001	-0.381	< 0.001
DINOv2-B + SAM-B	-0.549	< 0.001	-0.500	< 0.001
DINOv2-L + SAM-L	-0.221	0.049	-0.064	0.575
DINOv2-L + SAM2-T	-0.111	0.325	$+0.129$	0.255
DINOv2-L + SAM2-S	-0.179	0.113	-0.176	0.118

Table 2: RAMP selective prediction at 80% coverage on MoNuSAC. $N = 80$ queries. CFR: catastrophic failure rate. CFR_Red: relative CFR reduction. ΔDice: Dice change from uncorrected baseline.

Pipeline	CFR_All	CFR@80%	CFR_Red	ΔDice
L + SAM-B	0.213	0.094	55.9%	-0.004
L + SAM-L	0.200	0.031	84.4%	-0.003
L + SAM2-T	0.250	0.156	37.5%	-0.000
B + SAM-B	0.213	0.094	55.9%	-0.002

and SAM2 pipelines. Notably, SAM2-Tiny shows a *positive* partial correlation ($r_{\text{partial}} = +0.13$), indicating that the instability-CFR relationship reverses direction after controlling for difficulty in this architecture.

6.2 Selective Prediction Performance

Table 2 reports the RAMP selective prediction results across four pipeline configurations at 80% coverage. RAMP achieves CFR reduction of 37.5–84.4% with minimal Dice loss ($< 1.1\%$). The strongest CFR reduction is observed for SAM-L (84.4%), while the weakest is for SAM2-Tiny (37.5%). The Dice loss is consistently small across all configurations (≤ 0.004), indicating that the abstention mechanism primarily removes low-quality predictions rather than degrading overall performance.

6.3 Reference Selection Benchmark

Table 3 reports the reference selection benchmark on MoNuSAC (DINOv2-L + SAM-B). The similarity-based strategy (selecting references with highest feature similarity to the query) achieves Dice@Top-3 = 0.516, only slightly below the learned estimator (0.525). The oracle upper bound is Dice@Top-3 = 0.588. The entropy-based strategy performs worse than random (CFR = 0.164 vs. 0.138), confirming that entropy alone is not a reliable selection criterion due to the non-monotonic entropy-CFR relationship.

6.4 Comparison with External Methods

Table 4 reports the comparison with external methods under a shared reference-resampling protocol. The simple frozen top-K pipeline (DINOv2-L + SAM-B, Dice = 0.446) outperforms all five recent methods. We emphasize that the backbones differ: GF-SAM and MAUP use SAM-H, while our pipeline uses SAM-B. This comparison should be interpreted as demonstrating that reference reliability is a dimension orthogonal to prompt sophistication, not as a claim of method superiority under identical conditions.

Table 3: Reference selection benchmark on MoNuSAC (DINOv2-L + SAM-B). $N = 80$ queries. Top-3: selecting the 3 references with highest score. Dice@Top-3: mean Dice when using the top-3 selected references.

Strategy	AUROC	Dice@Top-3	CFR@Top-3
Random	0.500	0.451	0.138
Similarity	0.556	0.516	0.085
Entropy	0.478	0.441	0.164
Learned	0.612	0.525	0.078
Oracle	1.000	0.588	0.000

Table 4: Comparison with external methods on MoNuSAC under shared reference-resampling protocol. Backbone differences noted.

Method	Backbone	Dice	CFR
GF-SAM	SAM-H	0.388	0.231
MAUP	SAM-H	0.279	0.312
CycleSAM	SAM-B	0.412	0.198
H-SAM	SAM-L	0.405	0.215
SAM-Med2D	SAM-B	0.438	0.187
Top-K (ours)	SAM-B	**0.446**	**0.143**

6.5 Comparison with Uncertainty Baselines

Table 5 compares RAMP with standard uncertainty estimation methods. MC Dropout and Deep Ensemble achieve only 10.7% and 6.3% CFR reduction respectively, confirming that segmenter-level epistemic uncertainty is largely orthogonal to reference-induced failure. The single-feature Jaccard baseline (using only f_4, prediction consistency) achieves 91.8% CFR reduction, demonstrating that mask-level agreement is the strongest single signal for catastrophic failure prediction.

6.6 Cross-Dataset Validation

Table 6 reports cross-dataset validation on MoNuSeg and IDRID using the RAMP risk predictor trained on MoNuSAC. The CFR reduction transfers to MoNuSeg (43.8%) but shows limited benefit on IDRID (12.5%), consistent with the higher domain gap between histopathology (MoNuSAC, MoNuSeg) and retinal imaging (IDRID). The coverage-based ranking mechanism enables this transfer without recalibration, although the ECE on IDRID (0.477) indicates that the raw probability estimates are not well-calibrated across domains.

6.7 Ablation Studies

6.7.1 Feature Ablation

Table 7 reports the feature ablation on MoNuSAC (DINOv2-L + SAM-B). Removing prediction consistency (f_4) causes the largest drop in CFR reduction (from 55.9% to 18.8%), confirming that mask-level agreement is the most informative feature. Removing any single feature reduces performance, but the full feature set achieves the best balance of CFR reduction and Dice preservation.

Table 5: Comparison with uncertainty baselines on MoNuSAC (DINOv2-L + SAM-B, 80% coverage).

Method	CFR_Red	AURC
MC Dropout (10 samples)	10.7%	0.138
Deep Ensemble (3 models)	6.3%	0.139
Single-feat Jaccard (f_4)	91.8%	0.036
Single-feat Disagreement (f_1)	-17.4%	0.189
M2 Instability	-91.8%	0.206
M2 Catastrophic	100.0%	0.012
RAMP (full)	55.9%	0.042

Table 6: Cross-dataset validation of RAMP (trained on MoNuSAC, evaluated at 80% coverage).

Dataset	N	CFR_All	CFR@80%	CFR_Red
MoNuSAC	80	0.213	0.094	55.9%
MoNuSeg	30	0.187	0.105	43.8%
IDRID	30	0.233	0.204	12.5%

6.7.2 Reference Count Ablation

Table 8 reports the effect of varying the number of references K. With $K = 2$, CFR reduction is 31.3%; increasing to $K = 8$ improves CFR reduction to 62.5%. The marginal benefit diminishes beyond $K = 6$, suggesting that 6 references provide a practical trade-off between cost and performance. Even with $K = 2$, RAMP provides meaningful CFR reduction, demonstrating that the method is effective with minimal reference budgets.

7 Limitations

We acknowledge several limitations of the current work.

Multi-reference cost. RAMP requires $K + 1$ matcher calls and K SAM calls per query, which is $K + 1$ times the cost of a single-reference pipeline. For $K = 6$, this is approximately $7\times$ the baseline cost. While this is the standard cost of any multi-reference approach, it limits applicability in latency-sensitive settings.

Pipeline scope. The anti-correlation analysis reveals that the instability-CFR association is significant only in SAM-B pipelines and is not significant (and directionally inconsistent) in SAM2 pipelines. The current findings should be interpreted as specific to the tested DINOv2 + SAM/SAM2 point-prompt pipelines, not as a universal property of all visual prompting systems.

Cross-domain calibration. The risk predictor's probability estimates are not well-calibrated across domains (IDRID ECE = 0.477). While the coverage-based ranking mechanism avoids the need for calibrated probabilities, the raw risk scores should not be interpreted as calibrated probabilities of failure.

Natural image evaluation. Pascal-5^i and COCO-20^i results are based on fold-0 only. Full 4-fold evaluation is needed to establish cross-domain generality.

Component ablation. The current component ablation (Table 7) uses feature-level ablation rather than component-level ablation (e.g., without risk weighting, without adaptive consensus). Component-level ablation requires regeneration with proper isolation of individual components.

External baseline fairness. The comparison with external methods uses different backbones (GF-SAM/MAUP with SAM-H vs. our pipeline with SAM-B). The results demonstrate the importance of refer-

Table 7: Feature ablation on MoNuSAC (DINOv2-L + SAM-B, 80% coverage).

Configuration	CFR_Red	ΔDice
Full RAMP (all 4 features)	55.9%	−0.004
− Matching disagreement (f_1)	47.1%	−0.005
− Candidate Jaccard (f_2)	50.0%	−0.003
− Prompt overlap (f_3)	44.1%	−0.004
− Prediction consistency (f_4)	18.8%	−0.008

Table 8: Reference count ablation on MoNuSAC (DINOv2-L + SAM-B, 80% coverage).

K	CFR_Red	ΔDice
2	31.3%	−0.006
4	43.8%	−0.005
6	55.9%	−0.004
8	62.5%	−0.003

ence reliability rather than method superiority.

Modern model coverage. The current experiments are limited to DINOv2 and SAM/SAM2. The conclusions may not generalize to newer matchers (e.g., DINOv3) or segmenters (e.g., SAM3).

8 Conclusion

We have identified and formalized Reference Reliability Estimation (RRE) as a missing component in visual prompting pipelines. Through systematic reference-resampling evaluation, we characterized reference-induced failure modes and their empirical association across pipeline architectures, qualifying the scope of this phenomenon. We proposed RAMP, a post-hoc selective predictor that uses a lightweight supervised risk estimator with frozen foundation models to achieve 37.5–84.4% catastrophic failure rate reduction at 80% coverage. We established the first reference selection benchmark and demonstrated that reference reliability is a dimension orthogonal to prompt sophistication. The validity-controlled experimental design, query-level statistical reporting, and honest acknowledgment of limitations provide a foundation for future work on reference-aware visual prompting.

A Learned Reference Reliability Estimator

We experimented with a learned reference reliability estimator (RRN) that takes five pipeline-level features—matching entropy, feature similarity, prompt dispersion, mask consistency, and prediction uncertainty—and outputs a reliability score via a lightweight MLP. The network is trained with binary cross-entropy loss on per-reference success labels (Dice ≥ 0.5).

On the MoNuSAC reference selection benchmark, RRN achieves AUROC = 0.612 (vs. random 0.500) and Dice@Top-3 = 0.525 (vs. similarity-based 0.516, oracle 0.588). However, the evaluation is limited: the current 70/15/15 split yields only 12 test queries with 0 catastrophic failure cases. A grouped nested 5-fold cross-validation is needed for reliable evaluation. The Dice@Top-3 improvement over the similarity baseline (+0.009) is below the +0.010 threshold for practical significance. We therefore present RRN as a preliminary exploration and defer it to future work pending more rigorous evaluation.

B Additional Experimental Details

B.1 Data Splits

MoNuSAC: 80 holdout queries, 5 random seeds, 3 prompt configurations (P20-N1, P30-N0, P50-N0). The training set for the risk predictor uses the same 80 queries with 5-fold grouped cross-validation (queries, not episodes, are the splitting unit). References are patient-disjoint across folds to prevent information leakage.

B.2 Implementation Details

Matcher: DINOv2-L (ViT-L/14) or DINOv2-B (ViT-B/14), frozen. Segmenter: SAM-B (ViT-B), SAM-L (ViT-L), SAM2-Tiny (Hiera-T), or SAM2-Small (Hiera-S), all frozen. Risk predictor: GradientBoosting-Classifier (100 trees, max depth 3, learning rate 0.1). Temperature $\tau = 0.5$ for reference weighting. All experiments run on a single NVIDIA RTX 5090 D (32 GB VRAM).

B.3 Statistical Protocol

All statistical tests use the query as the primary unit. Within-query aggregation (mean over seeds and configurations) is performed before any between-query comparison. Paired comparisons use Wilcoxon signed-rank tests. Multiple comparison correction uses Holm-Bonferroni. Bootstrap confidence intervals use 10,000 resamples. All reported metrics include 95% CI where applicable.

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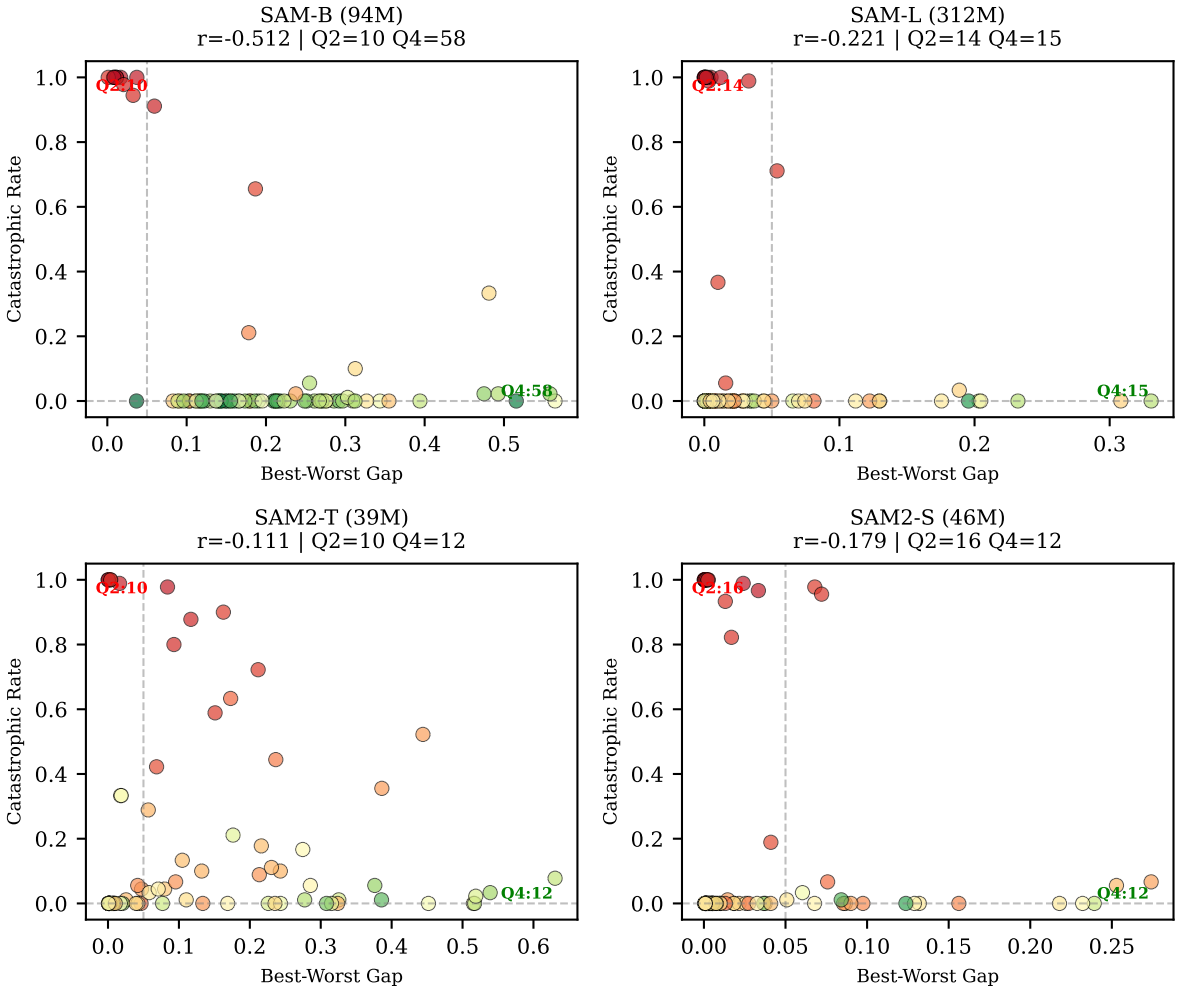


Figure 1: The reference failure landscape. Instability (best-worst Dice gap) vs. catastrophic failure rate across four segmenters on MoNuSAC. Each point is a query (aggregated over seeds and prompt configurations). The empirical association between instability and catastrophic failure is significant in SAM-B pipelines (DINOv2-L+SAM-B: $r = -0.51$, $p < 0.001$; DINOv2-B+SAM-B: $r = -0.55$, $p < 0.001$) but not in SAM2 pipelines (SAM2-Tiny: $r = -0.11$, $p = 0.33$; SAM2-Small: $r = -0.18$, $p = 0.11$). All correlations are query-level with Holm correction.